

Project Progress Report: GreenLawAI

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1. Overview

Since the project started, we have successfully moved from raw legal documents to a working "Legal Intelligence" system. We didn't want to build just another basic chatbot that guesses the law. Instead, we focused on building a system that actually reads the law before it speaks.

We've completed the heavy foundational work (**Phase 1**) and most of the intelligence layer (**Phase 2**).

2. Data Preprocessing

The biggest challenge was the source material. Legal documents like the Forest Act 1927 and KPK Forest Ordinance 2002 are messy, full of OCR errors, and structured in a way that regular AI can't handle.

How I processed the data (The 5 Phases):

1. **Phase 1: Raw Extraction:** I used specialized tools to pull the text out of the scanned PDFs. This step was tricky because of the side-notes and old-style formatting which often broke the sentences middle-way.
2. **Phase 2: Semantic Segmenting:** Instead of just cutting the text into random blocks, I built a smart "Segmenter." It reads the text and identifies where a **Chapter** begins, where a **Section** starts, and where a **Schedule** ends. This keeps the legal context alive.
3. **Phase 3: Hierarchy Mapping:** We converted the cleaned text into a structured JSON hierarchy. This means for every single paragraph, we know exactly which Law, Chapter, and Section it belongs to.

4. **Phase 4: Entity Extraction:** I wrote a script to find all the "Important Words" like tree species (Deodar), legal roles (Conservator), and penalties. These were used later to build our Knowledge Graph.
5. **Phase 5: Semantic Role Tagging:** This was my final cleanup. I tagged legal chunks as "Definitions," "Regulatory Rules," or "Penalties." This is what allows our AI to know it should look in a "Definitions" chunk when you ask "What is...?"

Why this matters: Most AI projects just dump text into a folder. By doing these 5 phases, we built a **Ground Truth** database where every word is tracked and organized.

3. The "Two-Brain" Search Engine (GraphRAG)

To make the search accurate, I decided to go beyond standard RAG. I implemented a **Hybrid Retrieval** system:

- **FAISS (Vector Database):** This searches for text that is semantically similar to the user's question.
- **Neo4j (Graph Database):** I built a Knowledge Graph where "Trees," "Penalties," and "Sections" are connected by nodes. If you ask about a specific tree, the graph knows exactly which penalty section is related to it.

Why this matters: It gives us the best of both retrieval i-e fuzzy search for natural language and structural search for legal accuracy.

4. Solving the Accuracy Problem

Our testing showed that the AI was getting confused by "circular definitions" (like saying "Timber is things made of timber"). I had to get into the backend to fix this.

My Fixes (The Guardrails):

1. **Expert Knowledge Overlays (EKO):** For high-stakes values like the Rs. 206,000 fine for Deodar, I hard-coded verified "Fact Chunks" that the system prioritizes above everything else.
2. **Intent Detection:** I wrote logic that analyzes the question before searching. If you're asking for a definition, the system "digs deeper" into the database (increasing the search depth from 5 to 100 matches) to find the Interpretations Clause.
3. **Statutory Priority:** I weighted the code to trust principal Acts and Ordinances more than minor administrative Rules.

5. Current System Status

The system is now operational on our local setup.

- **Interface:** Interface is built just for testing purpose using Streamlit, it is not the final. It's clean and shows the user the exact citations (Reasoning Logs) so they can verify the answer themselves.
- **Backend Link:** We are using a Hybrid-Cloud setup. The search happens on our local machine (for privacy), and the language processing (thinking) happens in Google Colab using a Llama 3 model.

6. What's Still Missing

While the core system is strong, we still have work to do:

- **Activity Monitoring:** We haven't yet built the logic to detect "illegal logging trends" from reports automatically.
- **Language Support:** The system is English-only right now; we still need to add the multilingual/Urdu support promised in the proposal.
- **Advanced GIS:** Integrating maps for forestry data is part of our final phase.

7. Conclusion

We have built the most difficult and basic part of the project: the **Legally Reliable Brain** which is also the Phase 1 of three phases. My current focus is refining and polishing the answers to be simpler and removing technical artifacts from the output.